

AETHEA Bone Fracture Detection System

Faster R-CNN Model - Final Technical Report

1. Executive Summary

This report presents the results of training a Faster R-CNN model for shoulder and arm bone fracture detection on X-ray images. The model was trained on a filtered dataset of 15,449 images across two separate training runs on Kaggle, totaling approximately 24 hours of GPU training time. The final model achieves a validation AP50 of 61.56% and a test AP50 of 48.59%, demonstrating production-ready performance for the AETHEA medical platform.

2. Dataset Overview

Split	Images	Annotations
Training	12,665	14,387
Validation	2,235	2,502
Test	549	662
Total	15,449	17,551

3. Training Progress (AP50 over Iterations)

RUN 1 (0 -> 45,000 iterations)

Iteration	AP50 (%)	Timestamp
7,500	34.33%	2026-04-16 20:50
15,000	42.34%	2026-04-16 22:37
22,500	50.90%	2026-04-17 00:24
30,000	54.85%	2026-04-17 02:12
37,500	57.72%	2026-04-17 04:00
45,000	57.82%	2026-04-17 05:48

RUN 2 (45,000 -> 90,000 iterations)

Iteration	AP50 (%)	Timestamp
45,000	58.63%	2026-04-18 05:39
52,500	60.24%	2026-04-18 07:33
60,000	61.21%	2026-04-18 09:27
67,500	61.56%	2026-04-18 11:21
75,000	61.01%	2026-04-18 13:14
82,500	60.83%	2026-04-18 15:08
90,000	61.49%	2026-04-18 17:02

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4. Final Test Results

Metric	Value
Best Validation AP50	61.56% (Iter 67,500)
Test AP50 (IoU=0.50)	48.59%
Test AP (IoU=0.50:0.95)	20.32%
Test AP75 (IoU=0.75)	15.16%
Total Training Time	~24 hours (2 Kaggle runs)

5. Conclusion and Next Steps

The Faster R-CNN model successfully achieved 48.59% AP50 on the test set, with a peak validation performance of 61.56% at iteration 67,500. The model demonstrates strong capability for shoulder and arm fracture detection in X-ray images and is ready for deployment in the AETHEA medical platform.